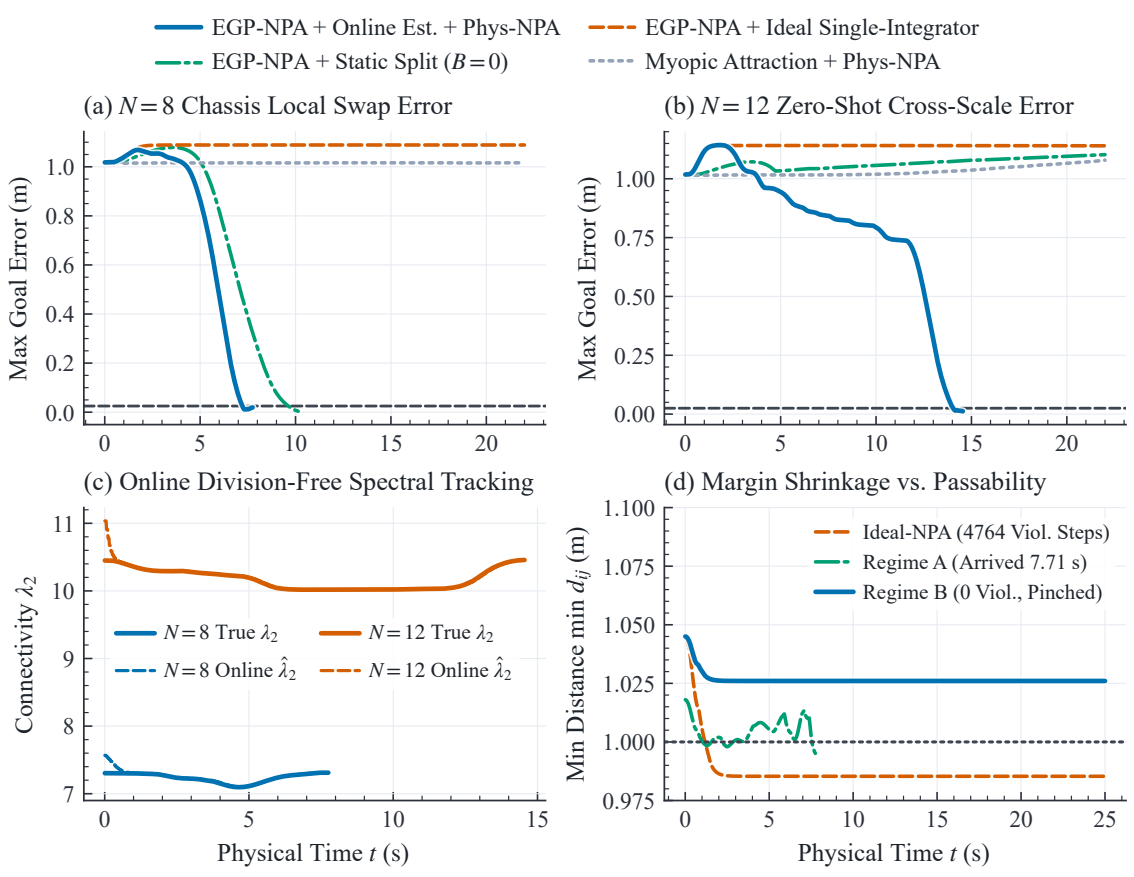

EGP-NPA + Online Est. + Phys-NPA
EGP-NPA + Ideal Single-Integrator
EGP-NPA + Static Split ($B=0$)
Myopic Attraction + Phys-NPA
(a) $N=8$ Chassis Local Swap Error
Max Goal Error (m)
(b) $N=12$ Zero-Shot Cross-Scale Error
Max Goal Error (m)
(c) Online Division-Free Spectral Tracking
Connectivity λ_2
$N=8$ True λ_2
$N=12$ True λ_2
$N=8$ Online $\hat{\lambda}_2$
$N=12$ Online $\hat{\lambda}_2$
Physical Time t (s)
(d) Margin Shrinkage vs. Passability
Min Distance min d_{ij} (m)
Ideal-NPA (4764 Viol. Steps)
Regime A (Arrived 7.71 s)
Regime B (0 Viol., Pinched)
Physical Time t (s)